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Single Coil Series Resonant Circuit for Pulsed Nuclear Resonance*

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This paper presents a description of and design optimization criteria for a series resonant single coil system suitable for use in pulsed nuclear magnetic or quadrupolar resonance. The design is adaptable to the frequency range 1–100 MHz, and permits both efficient use of transmitter power and signal reception with a high signal-to-noise ratio. An example of the design of a system to operate at 10 MHz is worked out and compared against the ideal case. An active damping circuit to speed the recovery following an rf pulse is also discussed.

INTRODUCTION

One of the most important parts of a pulsed nuclear resonance apparatus is the electrical system used to couple power from the transmitter into the transmitter coil and convert the precessing nuclear magnetization into a signal at the input of the receiver. The object of this paper is to describe such a system which has, with its precursors, been used in our laboratory over the last few years. We have found it to be capable of high performance and convenient to use.

The circuit described here has several desirable features. It uses the same series resonant sample coil for transmitting the rf pulse and receiving the nuclear induction signal. This feature makes the circuit suitable for pure nuclear quadrupolar resonance and facilitates its incorporation into pressure bombs, furnaces, etc., where entry of only one coil into that part of the apparatus is desirable. The design can be adapted to operate over a wide range of frequencies in practical circuits. (It must, however, be tuned to each frequency used; it is not a broadband system.) We have used it over the range 1–20 MHz, and believe its operation can be extended to 100 MHz or beyond. It permits both efficient power matching of the transmitter to the sample coil during rf pulsing and efficient coupling of the same coil to the signal amplifier during the nuclear induction signal. Adjustment of the circuit is simple. It is limited to ordinary tuning operations. Protection of the signal amplifier during a transmitter pulse is accomplished with diodes: There is no reliance on the balance of a bridge circuit to obtain this function. Finally, the circuit employs a simple active damping circuit, which significantly reduces the recovery time following the removal of an rf pulse.

The system described here draws upon earlier work by Lowe and Tarr,¹ in that it is an adaptation of their scheme for using crossed diodes to separate the functions of transmitting and receiving within the sample coil. It is different in that it avoids the use of resonant sections of transmission line and the actual sample coil is part of a series resonant system.²

Our description of the circuit includes the following. In Sec. I a brief and simplified description of the essential features in the operation of the circuit is given. Section II contains a detailed description of the components and connections which make up the working circuit. Then, in Sec. III operation of the circuit during the rf pulse phase is described in detail. Design criteria are presented which indicate how to optimize the circuit with

respect to producing a large rf magnetic field in the sample volume. An example of the application of the design criteria at a frequency of 10 MHz is presented and evaluated. It is found that the rf magnetic field in this example is 80% of the maximum obtainable in an ideal system.

In Sec. IV a discussion of circuit design and performance during the signal reception phase is given. Detailed design criteria for optimizing the signal-to-noise ratio are presented. The example at 10 MHz is continued to signal reception and evaluated. A calculated signal-to-noise ratio 84% of that for an ideal receiver coil circuit is obtained. Factors affecting the recovery time of the coil following an rf pulse are discussed, and the operation and performance of the active damping circuit are covered.

Finally, the procedure for tuning the circuits is presented in Sec. V.

Some comment on the optimization conditions presented below is warranted here. The expressions to be minimized are, in some cases, rather complicated, especially when proper account of the constraints imposed by physically realizable circuit components are included. In addition to this, mutually contradictory solutions may be indicated in some cases. For these reasons, it is not possible to give a complete formulation of how to optimize the circuit for all possibilities. Instead, it is necessary for the user to decide which functions are of greatest importance and optimize them at the possible expense of compromising on some other property. Part of the reason for including a worked out example at 10 MHz has been to illustrate what design choices are reasonable in a particular situation.

I. SIMPLIFIED CIRCUIT DESCRIPTION

In this section we discuss a simplified version of the circuit in order to illustrate the major principles of its operation. The treatment is only qualitative; in particular, no real losses are included in the circuits. A more detailed and quantitative discussion in support of the points made here is given in the subsequent sections.

The essential parts of the circuit are shown in Fig. 1(a), wherein T represents the transmitter and R the receiver. The inductor L_2 is the combination transmitter-receiver coil, which is closely wound over the sample.

The diode pairs D_1 – D_2 and D_3 – D_4 play a crucial role in the circuit. To a good approximation, they appear as a short circuit to the larger amplitude transmitter pulses and an open circuit to the subsequent small free induction decay signal (FID) induced in L_2 .

For the purpose of this discussion, it is assumed that the reactance pairs L_1, C_1 ; L_2, C_2 ; and L_3, C_3 have each been adjusted to resonance at the operating frequency ω .

Now consider what happens during a transmitter pulse. The diode pairs become a short circuit, and the resultant equivalent circuit becomes that of Fig. 1(b). The series tuned combination L_2, C_2 represents a very low impedance in parallel with that of L_3 . Hence, the loading effect of L_3 on the rest of the circuit is ignored.

Two important effects are achieved during the transmitter pulse with this configuration. First, it is efficient at conveying power from the transmitter into the sample coil L_2 . This occurs because the circuit can be designed to present an impedance at point A that closely matches the conditions for maximum power transfer from the plate circuit of the transmitter. The resultant circulating current appears in L_2 . The second property of the circuit is that there is very little voltage fed through to the receiver. This occurs for two reasons. First, point E has a low impedance (L_2 and C_2 in series) to ground. Hence, the voltage produced there is never very large. Secondly, the diodes D_3 and D_4 represent a short circuit to ground, so that the voltage at E is dropped across L_3 before reaching the receiver.

After the decay of the transmitter pulse, the diodes D_1 and D_2 effectively disconnect the transmitter from point E, and the diodes D_3 and D_4 revert to an open circuited condition. The result of this action is the equivalent circuit of Fig. 1(c) for receiving the FID. In that circuit L_2, C_2, L_3 , and C_3 represent a composite series circuit tuned to the operating frequency. The FID couples an oscillating signal into the circuit via L_2 . It is received at point F, which has a high impedance to ground. This high impedance level is usually favorable for obtaining a good signal-to-noise ratio in contemporary receiver designs.

In summary, the circuit permits efficient transfer of power from the transmitter to the sample coil, affords good protection of the receiver from destructive overloads, and provides a high signal-to-noise ratio for observation of the FID.

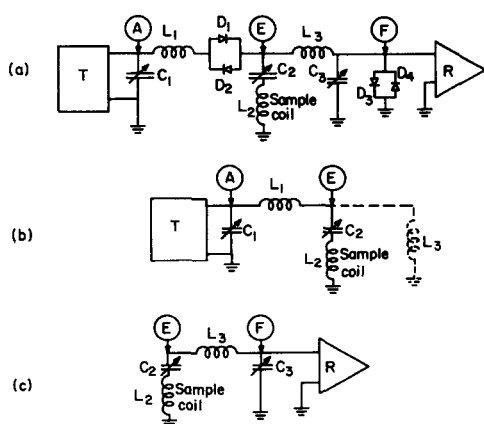


FIG. 1. Diagram of simplified circuit. (a) Full circuit; (b) equivalent transmitter circuit during the rf pulse; and (c) equivalent receiving circuit for the signal reception phase.

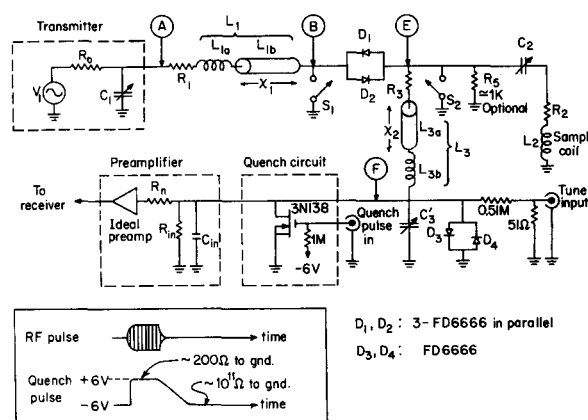


FIG. 2. Diagram of actual circuit as it is used. The inset shows the characteristics of the pulse driving the quench circuit and its time relationship with the rf pulse.

II. DESCRIPTION OF COMPONENTS

Now we turn to a detailed description of the configuration and components which make up a practical realization of the circuit. For this description refer to Fig. 2. The coil system consists of three tuned subsystems: the transmitter output circuit (R_1, L_1, C_1), the sample coil circuit (R_2, L_2, C_2), and the preamplifier input circuit ($R_3, R_{in}, C_{in}, C'_3, L_3$). In what follows, it will be assumed that each is tuned to the same frequency ω . (The actual tuning procedure is described below in Sec. V.) It will also be assumed that when each circuit is tuned it represents a high-Q circuit.

The transmitter we have used with these coils is similar to that described elsewhere.³ Its output characteristics are approximated by an internal voltage generator in series with an output resistance R_0 . The capacitance C_1 is an internal tuning capacitance in parallel with the output capacitance of the transmitter tubes. This capacitance is tuned by a series combination of a lumped inductance (L_{1a}) and a length x_1 of coaxial cable (L_{1b}).⁴ This cable is used to connect the transmitter to the sample coil circuit. As will be seen later, it is always terminated with an impedance small compared with its characteristic impedance Z_0 . The length is normally chosen to avoid multiples of $\frac{1}{4}\lambda$, where λ is the wavelength for the frequency used. Under these conditions, the cable acts mainly as a reactance $X = Z_0 \tan(2\pi x_1/\lambda)$.⁵ By choosing $0 \leq x \leq \lambda/8$, the cable can be chosen both to be a convenient length and represent an inductive element with inductance L_{1b} . The total inductance in the circuit is then $L_1 = L_{1a} + L_{1b}$. The resistor R_1 corresponds to the losses in the transmitter output circuit. Its origin is mainly from losses in L_{1a} and L_{1b} .

The switch S_1 is included to help with tuning. (It, and switch S_2 described below, are always open except for the tuning and Q measurement operations described in Sec. V.) It is left open during the transmission of power to L_2 . In our setup this switch is simply a shorted BNC plug which is connected into the circuit via a BNC T connector. The diodes D_1 and D_2 are as described on Fig. 2.

Now consider the sample coil circuit. The coil L_2 is wound closely on the sample. It is the combination receiving and transmitting coil. It is tuned by the series capacitance

C_2 . The resistance R_2 accounts for the losses in these components. Switch S_2 is used in the tuning procedure described in Sec. V. During rf pulsing it is left open. Its physical realization is identical to that of S_1 .

The representation of L_2 simply as an inductance requires further comment. In fact, the leads connecting C_2 to L_2 can have an appreciable effect on the circuit reactances. This is especially true, for example, in a setup where L_2 is at the bottom of a research Dewar, C_2 is at the top, and the two are connected by a coaxial cable. Under these circumstances the circuit analysis becomes somewhat more complicated than that presented here. In the interest of simplicity we shall restrict our present discussion to those cases where a simple inductive representation is sufficient. In practical terms this means either that the leads connecting C_2 and L_2 are short, or that the frequency is low enough that long leads are of little importance. In a later publication, we intend to describe the extension of this kind of circuit to the case where full account is taken of the lead length.

Finally, we consider the components of the preamplifier input circuit. It is coupled to the sample coil circuit via a lumped inductance L_{3b} and a section of coaxial line L_{3a} . The conditions of use for L_{3a} are identical to those already described for L_{1b} . Hence L_{3a} appears as an inductance in the circuit, and the total inductance L_3 is simply the sum $L_3 = L_{3a} + L_{3b}$. Normally, L_3 is chosen to be a convenient length for connecting the preamplifier to the sample coil circuit. In many of our setups the preamplifier is simply placed at C_2 and no length of cable is used. The resistance R_3 is the effective series resistance of L_3 . Inductance L_3 is tuned by capacitances C'_3 and C_{in} , which appear in parallel. For later reference, the preamplifier is represented as an ideal (infinite input impedance, noiseless) amplifier with shunt resistance and capacitance R_{in} and C_{in} , respectively. Its internally generated noise is that of thermal noise from an equivalent noise resistance R_n . The crossed diodes D3 and D4 are each an FD6666. Their function is described in Secs. III and IV.

The quench circuit is an active damping circuit to speed the recovery from ringing after an rf power pulse, as described, for example, in Ref. 3. The transistor looks like about 200 Ω to ground when the gate is pulsed to +6 V, and an open circuit when the gate resides at -6 V. The duration of the transition from +6 V to -6 V is chosen by trial and error to minimize the recovery time of the receiving system. A linear ramp recovery waveform is used to minimize additional ringing caused by pulsing the quench circuit. The relation of this waveform to the rf pulse is shown in the insert on Fig. 2.

Figure 2 also shows a tune input circuit. When the output of a CW signal generator is connected to the tune input connector, it injects a signal into the point F which can be used to tune the circuit and measure its Q, as described in Sec. V.

The resistor R_5 is used to lower the dc resistance to ground of points E and F when no other such path exists. This condition can occur, for example, when the rf coupling from the transmitter is by way of a capacitor. In this case,

unless R_5 is in the circuit, the dc return path is via the 0.51 M Ω resistor in the tune input circuit. When the resistance is so large, transients in the power supply feeding the output stage of the transmitter can couple to point F, and hence to the input of the preamplifier. This in turn modulates its gain and thereby degrades its performance. Addition of $R_5 \sim 1$ k Ω completely solves this problem in the system we use.³ It does not, however, affect the rf performance of the circuit, as it is connected at a point where the rf impedance to ground is only a few (~ 2 -5) ohms. In those cases where an alternative low impedance dc path to ground is provided at the output of the transmitter, R_5 can be omitted. In the following discussions of the rf performance of the circuit, no further mention will be made of R_5 .

III. TRANSMITTER PULSE CIRCUIT

In this section we describe in detail the operation of the system during a transmitter pulse. We shall proceed on the assumption that there are three major goals for the design: (a) protect the amplifier from destructive overloads, (b) ring down quickly enough at the end of the pulse to receive the FID, and (c) obtain as large an rf magnetic induction (B_1) as possible in the sample coil. It will be assumed that all problems associated with high voltage electrical breakdown during the transmitter pulse have already been solved for the components being used.

Protection of the amplifier has already been discussed in Sec. I. No further elaboration of this point is needed here except to say that in our experience the overload at the input of the preamplifier is typically about 0.5 V during an rf pulse.

The question of ring down time will be covered at more length near the end of this section. For the time being, we shall control this factor by assuming that the Q of the circuit has been specified.

Now we treat the question of maximizing B_1 during the rf pulse. The operation of the circuit during power pulse operation can be understood in terms of the equivalent circuit shown in Fig. 3. This circuit is derived from Fig. 2 by approximating both sets of crossed diodes as short circuits in the presence of high level signals and neglecting L_3 as discussed in Sec. I. (Switch S_1 is always open.)

The quantities considered as variables in this analysis are R_0 , C_1 , R_1 , L_1 , C_2 , R_2 , and L_2 . There are, however, numerous constraints on them. The transmitter output resistance R_0 is of course fixed by the transmitter design. All resonant circuits are tuned, i.e., $\omega = (L_1 C_1)^{-1/2} = (L_2 C_2)^{-1/2}$. The resonant Q of the composite transmitter circuit (Q_T) is assumed to be large,

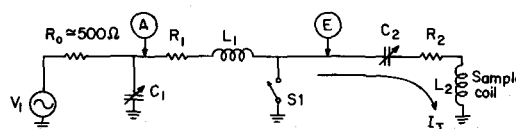


FIG. 3. Diagram of the equivalent transmitter circuit during an rf pulse.

$$Q_T \equiv \frac{\omega(L_1 + L_2)}{R_1 + R_2} \gg 1. \quad (1)$$

Finally, it will be assumed that variations in the sample coil L_2 are made only by changing the number of turns on the coil; no variation in the shape or volume (V) of this coil will be considered.

Now we turn to a calculation of B_1 . We use the usual convention that B_1 is one of the rotating components of the field, and is therefore one-half the magnitude of the oscillating field. Let I_T be the amplitude of the current circulating through the sample coil L_2 during an rf pulse. The instantaneous energy stored in the coil is $\frac{1}{2}L_2 I_T^2$. A good estimate of B_1 is obtained by assuming that half the total energy of the coil is stored in the magnetic energy enclosed by the volume of the coil. This leads to

$$B_1 = \left(\frac{\mu_0 L_2}{8V} \right)^{\frac{1}{2}} I_T, \quad (2)$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m, and a nonmagnetic sample is assumed. From elementary ac circuit theory, I_T is given by

$$I_T = \frac{-iV_1}{\omega L_1 + \omega C_1 R_0(R_1 + R_2) - i(R_1 + R_2)}. \quad (3)$$

We now assume that $(R_1 + R_2)$ is negligible compared to ωL_1 (high Q). It then follows that

$$I_T = \frac{-iV_1}{\omega L_1 + \omega R_0 C_1(R_1 + R_2)} \quad (4)$$

and

$$B_1 = \frac{V_1 \omega L_1 (\mu_0 L_2)^{\frac{1}{2}}}{(8V)^{\frac{1}{2}} [\omega^2 L_1^2 + R_0(R_1 + R_2)]}. \quad (5)$$

The factor i has been dropped, as we are interested in the magnitude of B_1 . In terms of Q_T , Eq. (5) becomes

$$B_1 = \frac{V_1 (\mu_0 L_2)^{\frac{1}{2}}}{(8V)^{\frac{1}{2}} [\omega L_1 + (R_0/Q_T)(1 + L_2/L_1)]}. \quad (6)$$

Variation of B_1 with respect to L_1 yields the extremal condition

$$L_1 = \left(\frac{L_2 R_0}{Q_T \omega} \right)^{\frac{1}{2}}, \quad (7)$$

and therefore

$$B_1 = \frac{V_1 (Q_T \mu_0)^{\frac{1}{2}}}{(32V \omega R_0)^{\frac{1}{2}} [1 + (R_0/4Q_T \omega L_2)^{\frac{1}{2}}]}. \quad (8)$$

From Eq. (7) it is seen that variation of L_2 to maximize B_1 leads to $L_2 \rightarrow \infty$, a physically unrealistic limit. It can also be seen, however, that if the condition

$$R_0 \leq \omega L_2 Q_T \leq 10 R_0 \quad (9)$$

is met, B_1 will not be less than two-thirds of its maximum possible value. We therefore choose Eq. (9) as a practical criterion for determining L_2 . In addition, for the value $R_0 \approx 500 \Omega$ this range turns out to be a convenient design

choice. Once L_2 has been chosen, the value for L_1 is obtained from Eq. (7).

The value of Q_T to be chosen can be obtained from the relationship

$$\omega \tau = 2Q_T, \quad (10)$$

where τ is the time for the circuit to ring down in amplitude by the factor e^{-1} . The output resistance of the transmitter does not appear in this expression, as R_0 is normally very large after the transmitter driving pulse has been terminated. The actual value chosen for Q_T will depend on the requirements of the experiment in question or what is physically realizable with circuit components.

Equations (7), (9), and (10) constitute the main design criteria for the components of the transmitter pulse circuit. A typical set of values would be $\omega = 6 \times 10^7 \text{ sec}^{-1}$, $R_0 = 500 \Omega$, and $\tau = 3 \times 10^{-7} \text{ sec}$. The choice $\omega L_2 Q_T = 4 R_0$ makes B_1 80% of the maximum obtainable [Eq. (8)]. The design criteria then specify $Q_T = 9$, $L_2 = 3.7 \mu\text{H}$, and $L_1 = 1.8 \mu\text{H}$. The tuning condition yields $C_2 = 75 \text{ pF}$ and $C_1 = 150 \text{ pF}$. The value of Q_T requires $R_1 + R_2 = 37 \Omega$. This would ordinarily be achieved by adding whatever discrete resistor is required at R_1 . At this point it is instructive to estimate the actual value of B_1 expected of the example just given. First, it is useful to recast Eqs. (7) and (8) into the form

$$B_1 = \left(\frac{\mu_0 P \tau}{8V} \right)^{\frac{1}{2}} \left[1 + \frac{L_1}{2L_2} \right]^{-1}, \quad (11)$$

where use has been made of Eq. (10), and P is the transmitter power output capability into a matched resistance load. P is related to V_1 and R_0 by

$$V_1 = (8PR_0)^{\frac{1}{2}}. \quad (12)$$

Now let $P = 5 \text{ kW}$ and $V = 1 \text{ cm}^3$. The resultant value for B_1 is then $1.2 \times 10^{-2} \text{ T}$ (120 Oe).

One additional comment should be made regarding the ring-down time [Eq. (10)]. There exist nonlinear circuits⁶ whose object is to permit a larger Q (and hence larger B_1) than that indicated by Eq. (10). We have restricted our discussion here to linear passive circuits in order to keep the problem from becoming too difficult. It is our experience that the design criteria given here for linear circuits are usually sufficient for handling the more complicated nonlinear circuits as well, except that a larger value of Q_T than that of Eq. (10) can be used.

Finally, we point out that although the discussion given here is aimed at a system which uses the same coil for transmitting B_1 and receiving the FID, it can also be applied to a crossed coil system as well. In this case, one would simply use L_2 as the separate transmitter coil and omit the connection to the receiving system at E (Fig. 2). The diodes D_1 and D_2 can be left in. They have the beneficial effect of rejecting noise signals passing from the transmitter to L_2 in the absence of an rf pulse, and yet cause a negligible degradation of the rf pulse.

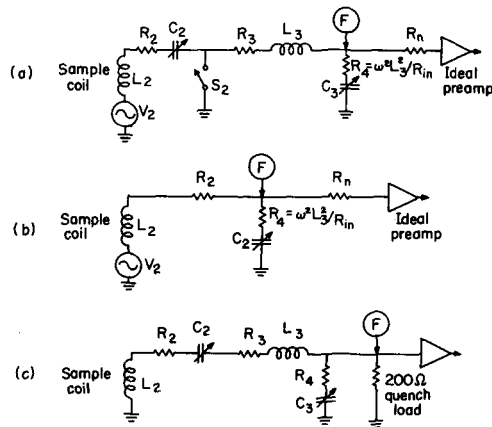


FIG. 4. (a) Diagram of the equivalent signal receiving circuit for the complete coil system. (b) Diagram of the equivalent signal receiving circuit for a simple parallel tuned receiver coil. (c) Diagram of the equivalent signal receiving circuit during operation of the quench circuits.

IV. SIGNAL RECEPTION CIRCUIT

In this section we consider the factors which affect the design of the coil system for efficient reception of the FID. The first consideration covered here is to approach the largest signal-to-noise ratio possible within the limitations of real physical components. Later, some attention is paid to the problem of obtaining fast recovery of the reception circuits from the shock of the rf transmitter pulse. As in Sec. IV, it will be assumed that all variations of parameters will be made relative to a constant sample coil geometry.

The equivalent circuit used for this analysis is shown in Fig. 4(a). It is derived from Fig. 2 by treating all pairs of crossed diodes as an open circuit, along with the following considerations. Because of its large impedance, the loading effect of the tune input circuit is ignored. It is assumed that the quench circuit is not activated, and therefore it represents an infinite impedance. The capacitance C_3 in Fig. 4 is the sum of C'_3 and C_{in} in Fig. 2. Finally the parallel combination of R_{in} and C_3 is replaced by its series equivalent R_4 and C_3 , where

$$R_4 = \omega^2 L_3^2 / R_{in}. \quad (13)$$

The assumption of high-Q circuits has been used in Eq. (13) and will be continued below where appropriate. Finally, it is assumed that the conditions of tuning have been achieved for the resonant pairs L_2, C_2 , and L_3, C_3 ; i.e., $\omega L_2 = (\omega C_2)^{-1}$ and $\omega L_3 = (\omega C_3)^{-1}$.

The generator V_2 represents the voltage induced in the sample coil L_2 by the nuclear induction. We assume for the time being that the precessing nuclear magnetization has a steady state value M_0 and calculate V_2 as follows (mks units are used). For the single layer solenoids typically used as NMR receiving coils⁷ it can be shown that the relation between L_2 , the number of turns (n), and the coil dimensions is

$$n = \left[\left(\frac{L_2 l}{\mu_0 A} \right) \left(1 + 0.9 \frac{r}{l} \right) \right]^{\frac{1}{2}}. \quad (14)$$

where l is the length of the coil, A is the cross sectional area, and r its radius. Using Faraday's law, the magnitude of the voltage V_2 , is found to be

$$V_2 = n \omega \mu_0 M_0 A f, \quad (15)$$

where f is a geometric filling factor ($f=1$ for the induction due to precessing magnetization from an infinite cylinder coaxial with the sample coil and with the same cross sectional area A). Combining Eqs. (14) and (15) gives

$$V_2 = f \omega M_0 \left[L_2 \mu_0 V \left(1 + 0.9 \frac{r}{l} \right) \right]^{\frac{1}{2}} \quad (16)$$

(V is the volume of the coil).

Now we calculate the amplitude of signal appearing at the preamp (S). At resonance, the current in the loop (I_R) is given by

$$I_R = \frac{V_2}{R_2 + R_3 + R_4}. \quad (17)$$

Then S is simply the product of I_R and the magnitude of the series impedance of R_4 and C_3 ,

$$S = \frac{V_2 \omega L_3}{R_2 + R_3 + R_4} \left(1 + \frac{\omega^2 L_3^2}{R_{in}^2} \right)^{\frac{1}{2}}. \quad (18)$$

Use of the high-Q condition $\omega L_3 \ll R_{in}$ and Eq. (16) then yields

$$S = f M_0 \left[\mu_0 \omega V \left(1 + 0.9 \frac{r}{l} \right) \right]^{\frac{1}{2}} \left[\frac{\omega^{\frac{1}{2}} L_2^{\frac{1}{2}} \omega L_3}{R_2 + R_3 + R_4} \right]. \quad (19)$$

Our next task is to calculate the noise voltage N appearing at the input of the preamplifier. We shall assume that the sources of noise are thermal noise due to the dissipative elements of the signal reception circuit plus the noise generated in the first stage of the amplification chain. In addition, it will be assumed that the over-all system noise bandwidth B (in Hz) is much less than the bandwidth of the signal reception circuit, so that only the noise generated in this circuit at the resonance frequency ω need be considered. In the model used here the noise originates in each of the resistors (R_2, R_3, R_4 , and R_n) according to the Nyquist formula,

$$N_i^2 = 4kT_i R_i B, \quad (20)$$

where k is Boltzmann's constant and T_i is the temperature of the i th resistor (in K). Since noise is random, the noise voltage (N_R) around the loop traveled by I_R is obtained by adding the squares of the individual voltage sources,

$$N_R^2 = N_2^2 + N_3^2 + N_4^2 = 4kB(T_S R_2 + T_L R_3 + T_L R_4). \quad (21)$$

Here we have assumed that R_2 is at the sample temperature T_S and R_3 and R_4 are at the laboratory temperature T_L . As with the signal, the noise voltage appearing at point F (N_F) is obtained by calculating the effect of the current

in the impedance formed by R_4 and C_3 ,

$$N_F^2 = \frac{4k B \omega^2 L_3^2 (R_2 T_S + R_3 T_L + R_4 T_L)}{(R_2 + R_3 + R_4)^2}. \quad (22)$$

To this must be added the noise from R_n , which results in the total rms noise voltage at the input of the ideal system (N),

$$N = \left(\frac{\omega L_3}{R_2 + R_3 + R_4} \right) (4k T_L B)^{\frac{1}{2}} \times \left[\frac{T_S}{R_2} + \frac{R_3}{R_2 + R_3 + R_4} + \frac{R_n}{\omega^2 L_3^2} (R_2 + R_3 + R_4)^2 \right]^{\frac{1}{2}}. \quad (23)$$

Combining Eqs. (19) and (23) yields the following for the system signal-to-noise ratio (S/N):

$$\frac{S}{N} = \frac{f M_0}{2} \left[\frac{\mu_0 \omega V}{k T_L B} \left(1 + 0.9 \frac{r}{l} \right) \right]^{\frac{1}{2}} \left(\frac{1}{J} \right)^{\frac{1}{2}}, \quad (24)$$

where

$$J = \frac{R_2 T_S}{\omega L_2 T_L} + \frac{R_3}{\omega L_2} + \frac{R_4}{\omega L_2} + \frac{R_n}{\omega^3 L_2 L_3^2} (R_2 + R_3 + R_4)^2. \quad (25)$$

In Eqs. (24) and (25) factors have been separated into two groups: those related to impedances of the circuit [Eq. (25)], and those related to details of the sample, such as nuclear species (M_0), resonant frequency (ω), sample volume and geometry, etc. Although Eq. (24) is useful as a guide for optimizing the sample-related factors, we shall not consider that question here, as the purpose of this paper is to discuss the design and function of an electrical circuit. We limit our treatment to those factors directly related to circuit values.

Optimizing S/N therefore constitutes seeking the smallest possible value for J . For the purpose of discussing the ways of minimizing J it is convenient to change the variables appearing in Eq. (25). The appropriate transformations are

$$Q_2 = \frac{\omega L_2}{R_2}, \quad (26)$$

$$Q_3 = \frac{\omega L_3}{R_3}, \quad (27)$$

$$Q_L = \frac{\omega L_3}{R_3 + R_4}, \quad (28)$$

$$Q_R = \frac{\omega (L_2 + L_3)}{R_2 + R_3 + R_4}. \quad (29)$$

The quantity Q_2 is the Q of the sample coil, Q_3 is the Q of inductance L_3 , Q_R is the Q of the full receiving circuit, and Q_L is the Q of the resonant circuit composed of L_3 , C_3 , R_3 , and R_4 . The latter two values of Q are what would be measured using the tuning procedure of Sec. V with switch S_2 open and closed, respectively.

By using these definitions and Eq. (13), J can be expressed as

$$J = \frac{1}{Q_2} \frac{T_S}{T_L} + \frac{L_3}{L_2} \left(\frac{1}{Q_3} + \frac{\omega L_3}{R_{in}} \right) + \frac{R_n}{\omega L_2} \left(\frac{L_2}{L_3 Q_2} + \frac{1}{Q_3} + \frac{\omega L_3}{R_{in}} \right)^2, \quad (30)$$

or

$$J = \frac{T_S}{Q_R T_L} \left(1 + \frac{L_3}{L_2} \right) + \frac{L_3}{L_2 Q_L} \left(1 - \frac{T_S}{T_L} \right) + \frac{R_n}{\omega L_2 Q_R^2} \left(1 + \frac{L_2}{L_3} \right)^2. \quad (31)$$

For the purposes of evaluating the signal reception circuit, it is useful to compare its performance with that of a simple conventional tuned circuit using a similar sample coil. Such a circuit, with the same preamplifier, is shown in Fig. 4(b). The same line of analysis which led to Eqs. (24), (30), and (31) gives for S/N

$$\frac{S}{N} = \frac{f M_0}{2} \left[\frac{\mu_0 \omega V}{k T_L B} \left(1 + 0.9 \frac{r}{l} \right) \right]^{\frac{1}{2}} \left(\frac{1}{K} \right)^{\frac{1}{2}}, \quad (32)$$

where

$$K = \frac{1}{Q_2} \frac{T_S}{T_L} + \frac{\omega L_2}{R_{in}} + \frac{R_n}{\omega L_2} \left(\frac{1}{Q_2} + \frac{\omega L_2}{R_{in}} \right)^2, \quad (33)$$

and with

$$Q'_L = \frac{\omega L_2}{R_2 + R_4}, \quad (34)$$

$$K = \frac{1}{Q'_L} \left[1 + \left(\frac{T_S}{T_L} - 1 \right) \frac{Q'_L}{Q_2} + \frac{R_n}{\omega L_2 Q'_L} \right]. \quad (35)$$

Equations (32)–(35) can, of course, be used as a basis for optimizing a simple receiver coil when a conventional crossed coil system is being employed.

Because of the many different experimental situations which can occur and the complexity of Eqs. (30) and (31), it is not practical to discuss a purely mathematical solution to minimizing J . Instead, we shall make some general comments and follow them with a specific example which corresponds to a commonly met situation. It will be assumed that R_n and R_{in} are known at the frequency ω .

First, there is the question of whether to use Eq. (30) or Eq. (31). It is usually most convenient to employ the former if the Q's of the individual inductors are known and it is not necessary to limit Q_L because the FID lifetime is short. Under these circumstances it is clear by inspection of Eq. (30) that Q_2 and Q_3 should be made as large as possible, other quantities being equal. On the other hand, when Q_L must be limited by the FID lifetime, Eq. (31) is usually more appropriate. It also has the advantage of being expressed in terms of Q_L and Q_R , which are easily measured in the circuit (see Sec. V).

It is our experience that having chosen between Eq. (30) and Eq. (31), the best way to proceed is by trial and error. A trial set of reasonable values is proposed for J consistent with the requirements of tuning. The various contributions to J are then computed. There will usually be a small number of terms which dominate. The component values are then readjusted so that the predominant variable terms

in J give comparable contributions. This set of values usually represents about the best that can be done. Furthermore, moderate variations of these choices about the best values usually produce negligible reductions in performance; hence, a rather crude optimization is sufficient for most cases.

Now we illustrate this approach with an example, which continues from that given in Sec. III ($L_2 = 3.7 \mu\text{H}$). Assume $T_S = T_L$, and the highest Q obtainable can be used. Under these conditions the quantity to be minimized is Eq. (30). Let $\omega = 6 \times 10^7 \text{ sec}^{-1}$, $R_{in} = 4 \times 10^5 \Omega$, and $R_n = 250 \Omega$. (The latter values are appropriate for current low noise, high frequency transistors such as the 3N200 or 3N204.) Reasonable values for Q_2 and Q_3 are 50 and 100, respectively. (The restricted geometry of the sample coil often dictates a smaller Q for it than for L_3 .) The only quantity left to vary is L_3 . The optimum corresponds approximately to $L_3 = \frac{1}{2} L_2 = 1.85 \mu\text{H}$. The resultant value for J is $J = 1.40 Q_2^{-1}$, which yields S/N proportional to $[Eq. (24)] J^{-1} = 0.84 Q_2^{\frac{1}{2}}$. For completeness, we note that the tuning condition gives $C_2 = 75 \text{ pF}$ and $C_3 = 150 \text{ pF}$. It is worth remarking that if we had allowed variation of L_2 in this example, S/N could have been improved by only about 10% before L_2 and L_3 became unreasonably large. Also, note that the best that could ever be done when $T_S = T_L$ is $J^{-1} = 1.00 Q_2^{\frac{1}{2}}$. Hence, the circuit solution given here missed the ideal case for this circuit by only 16%.

Now let us compare this example with what would be obtained using the simpler circuit of Fig. 4(b) [Eq. (33)]. The same conditions on T_S and Q_2 as before are to be used. Variation of L_2 minimizes K at a value which is essentially indistinguishable from the previous used value $L_2 = 3.7 \mu\text{H}$. In this case S/N corresponds to $K^{-1} = 0.97 Q_2^{\frac{1}{2}}$.

Comparison of this value with that obtained for the more complicated circuit ($J^{-1} = 0.84 Q_2^{\frac{1}{2}}$) shows a relative loss in S/N by the factor 0.86. This reduction is to be expected, as there are additional components to cause circuit losses and noise.

The additional circuit complexity plus the 14% relative loss in S/N are the price paid in this example for obtaining the benefits of the circuit described here.

Now we conclude this section with a discussion of the recovery time for the signal receiving circuit. We shall concentrate on the circuit already treated as an example. There are three stages to be considered in the recovery of the coil system (a) the ringdown from the kilovolt levels of the transmitter pulse to the 0.5 V level of the diode conduction threshold, (b) the ringdown from 0.5 V to FID voltage levels of $\sim 5 \mu\text{V}$ using the quench circuit to lower the Q of the signal receiving circuit, and (c) the ringup of the circuit to its steady-state level after removal of the quench restores the Q of the receiving circuit. The first step has already been covered in Sec. III. Here, we consider the second and third points. It will be assumed that the switching waveform operating the quench is slow enough that no spurious oscillations are introduced, and that there are no complications from effects such as magnetoacoustic ringing.³

The equivalent circuit with the quench circuit operating is shown in Fig. 4(c). It is derived from Fig. 2 and Fig. 4(a). (Switch S_2 is assumed to be open.)

Now consider the phase going from 0.5 V to $5 \mu\text{V}$. This corresponds to about 12 decay time constants τ_d . From elementary circuit theory, the time constant for the decay of oscillations in the circuit is approximately $\tau_d = 2 RC$, where $R \approx 200 \Omega$ and $C \approx 10^{-10} \text{ F}$; $\tau_d = 4 \times 10^{-8} \text{ sec}$. To cover the range from 0.5 V to $5 \mu\text{V}$ therefore takes about $0.5 \mu\text{sec}$, which is quite short. For the final phase, the time constant for ringing up τ_u is $\tau_u = 2Q_R^{-1}$. At $\omega = 6 \times 10^7 \text{ sec}^{-1}$ and $Q_R = 50$, $\tau_u = 1.7 \mu\text{sec}$. Recovery to within 2% of the steady state value takes $4\tau_u \approx 7 \mu\text{sec}$. This illustrates that a major portion of the recovery is the time it takes for the circuit to recover following removal of the quench pulse. If the high voltage phase of the ring-down is adjusted to a value of $\sim 2.5 \mu\text{sec}$, the complete recovery of the circuit is about $10 \mu\text{sec}$. This value is typical of circuits we have used following the designs presented here. Still shorter times can, of course, be obtained at the expense of lower Q 's.

It is useful to examine what behavior would obtain if no quench were available. In that case $\tau_d = \tau_u = 1.7 \mu\text{sec}$. The full ring-down time would then be $12 \tau_d = 20 \mu\text{sec}$. This value corresponds to the behavior we observe when no quench is used. From this example it is seen that the quench provides an important improvement in the recovery time of the circuit.

V. TUNING PROCEDURE

There are three sets of reactances to be tuned in the circuit, L_1 and C_1 , L_2 and C_2 , and L_3 and C_3 . These tunings are easily accomplished as follows. First, the pair L_1 , C_1 is tuned by closing switch S_1 and adjusting C_1 for a maximum amplitude rf pulse as measured at point A of Fig. 2.

Next, the pair L_3 , C_3 is tuned. For this operation, S_2 is closed and a low level rf signal is introduced at the tune input (Fig. 2). Because of the $0.51 \text{ M}\Omega$ resistor feeding point F, a constant current is injected into L_3 and C_3 , which constitute a parallel RLC circuit when S_2 is closed. Tuning is accomplished by adjusting C_3 for a maximum amplitude signal at point F. This voltage is monitored by measuring the rf amplitude at the output of the preamplifier or receiver.

Once C_3 has been adjusted, Q_1 [Eq. (28)] can be easily measured. First, the frequency of the applied rf signal is varied while keeping its amplitude constant. Then, the upper and lower frequencies ω_1 and ω_2 at which the voltage at F has dropped to $1/\sqrt{2}$ of its maximum value are measured, and Q_L is computed according to

$$Q_L = \omega(\omega_1 - \omega_2)^{-1}. \quad (36)$$

The final tuning operation is that of L_2 and C_2 . Switches S_1 and S_2 are opened, so that the entire signal reception circuit is fed by the rf signal generator [Fig. 4(a)]. It is easily shown that since L_3 and C_3 are already tuned, L_2 and C_2 can simply be tuned by tuning the entire circuit. This is accomplished by adjusting C_2 to give a voltage

maximum at F , as described above for tuning C_3 . Also, if desired, Q_R [Eq. (29)] can now be measured by following the same prescription given above for Q_L . These steps complete the tuning of the entire circuit.

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¹I. J. Lowe and C. E. Tarr, J. Phys. E **1**, 320 (1968).

²An earlier adaptation of the ideas in Ref. 1 has also been developed in our laboratory. It uses quarter wavelength cables and a series resonant sample circuit. It is described briefly in A. L. Kerlin, Ph.D. thesis, University of California, Los Angeles (1972). The circuit described here is a modification of the earlier circuit in which all or part of the quarter wavelength lines are replaced with lumped elements and certain other components are eliminated. It is now used in preference to the former as it is much more flexible and permits better optimization of its performance.

³W. G. Clark, Rev. Sci. Instrum. **35**, 316 (1964).

⁴We commonly use RG-58 (50 Ω) cable in this application.

Most other cables would be equally suitable, provided their losses are not too large.

⁵Ronald W. P. King, in *Transmission-line Theory* (Dover, New York, 1965), p. 134. Strictly speaking, the inductive contribution of the line should be considered as frequency dependent when calculating the frequency response and Q of circuits containing transmission line elements. In addition, there are modifications to the transient response of the circuit. For the short lengths of transmission line given by $0 \leq x \leq \lambda/8$, these effects are small and will be ignored. At very high frequencies, a section of cable for which $0 \leq x \leq \lambda/8$ may be inconveniently short. By going to $\lambda/2 \leq x \leq 5\lambda/8$, etc., the cable can be made longer and still satisfy the tuning requirements presented in this paper. There will, however, be changes in the transient and frequency responses of the circuits. These changes are beyond the scope of this paper. Nevertheless, we believe that from a practical point of view, these changes will not be important in most applications.

⁶J. J. Spokas, Rev. Sci. Instrum. **36**, 1436 (1965); and G. L. Samuelson and D. C. Ailon, Rev. Sci. Instrum. **41**, 1601 (1970).

⁷See, *Reference Data for Radio Engineers*, edited by H. P. Westman (International Telephone and Telegraph Corp., New York, 1956), p. 112.